

Yiyang Wu

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Skills

Languages & Tools: Python, C/C++, SQL, Bash, Docker, Git, Linux, Slurm, AWS, WandB

Frameworks: PyTorch, Transformers, Hugging Face, vLLM, veRL, LLaMA Factory, FAISS, NumPy

Focus: LLM Post-Training, Agentic RL, CoT Supervision, Coding Agents, Multimodal Learning

Education

Carnegie Mellon University

M.S. in Intelligent Information Systems

Pittsburgh, PA

Expected Dec 2026

University of Southern California

B.S. in Computer Science (GPA: 3.98/4.00)

Los Angeles, CA

May 2025

Research Experience

CMU CX Research Group

Research Assistant

Pittsburgh, PA

Aug 2025 – Present

RL Post-training and Mid-training for LLMs

- Trained multiple **Qwen3-family base** and **instruct** checkpoints on the **DeepScaleR** dataset using the **GRPO** algorithm in the **veRL** framework, comparing post-training performance, **token-level entropy**, and training dynamics.
- Generated and analyzed synthetic **CoT** traces from **DeepSeek-R1**, then built a **FAISS**-based pipeline over **DCLM** to construct targeted mid-training data and run small-scale continual pretraining experiments.

Agent Test-time Scaling and Process Reward Design in Agentic RL

- Studied **test-time scaling** for **general-use agents** by increasing attempts on benchmarks like MBPP, WebVoyager, Terminal-Bench, Tau2-Bench, analyzing how **best-of- k** improves over **pass@1** on long-horizon tasks and identifying a persistent **verification gap** between **pass@ k** and self-selection ability.
- Built and published **Coding Agent Trajectory Viewer**, a **VS Code** extension that captures **Claude Code–Anthropic** API calls and visualizes response/action trajectories in an interactive webview for long-horizon agent analysis.
- Exploring **process reward modeling** for coding agents using hindsight reasoning and metadata from API captures.

USC LIME Lab

Research Assistant

Los Angeles, CA

Aug 2024 – Dec 2025

AdaRFT: Adaptive RL for Efficient Math Reasoning

- Designed **AdaRFT**, an adaptive curriculum method for **sample-efficient reinforcement learning** that updates task difficulty from reward signals during **PPO/GRPO** training using **veRL** on **Qwen2.5-1.5B** and **Qwen2.5-7B**.
- Estimated problem difficulty with **pass@128**, compared it against rubric-based **LLM-as-a-judge** evaluation, and found empirical support for **pass@128** as a strong difficulty metric.
- Achieved **2× faster convergence** and **3–4% accuracy gains** on competition-level math benchmarks; work accepted at the **NeurIPS 2025 Efficient Reasoning Workshop** (arXiv:2504.05520) with open-source code and dataset release.

CoT Supervision and SFT for Long-context QA

- Generated and inspected GPT-synthesized **CoT** supervision from **LongBench** answers, curating **SFT** pairs of question, synthetic reasoning trace, and answer.
- Trained **Qwen3-4B** in **LLaMA Factory** with continual pretraining on original **LongBench** long-context data followed by synthetic-CoT **SFT** to study whether **parametric memory** improves long-context QA performance.

Selected Projects

SemZoom: Semantic-Guided Zooming for High-Resolution VQA

Research Developer

Pittsburgh, PA

Jan 2026 – Present

- Designed **SemZoom** for **V*** and **HR-Bench 4K/8K**, representing each patch and its metadata as a node in a priority-queue tree search to localize **tiny targets** (**V*** average target area: **0.136%** of the image).
- Implemented and analyzed **caption-based confidence scoring**, **object-list prompting**, **semantic-crop fallback** using **GPT-5.4**, and **noise-reduced merge** with low-information gray padding for unselected patches; analysis showed image-preserving semantic guidance worked better than text-only compression.
- Ran weekly ablations and error analysis mainly on **V***, plus evaluation on **HR-Bench 4K/8K**; the best setting reached **93.2%** total accuracy on **V***, a **2%** gain over the baseline tree-search algorithm.

Workflow Memory for Coding Agents

Research Developer

Pittsburgh, PA

Sep 2025 – Dec 2025

- Ran **OpenHands + Kimi K2** on **SWE-bench** and analyzed failures to identify dominant failure modes.
- Built a **workflow-memory** layer on top of the pipeline, converting successful **ReAct** repair trajectories into concise, reusable workflows via LLM-based cleaning, compression, and abstraction.
- Designed **offline** and **online** workflow-memory settings: offline injects a static workflow library and improved the workflow-free baseline by **2%**, while online incrementally compresses successful trajectories during inference to update memory.